PatientTriage.ai

“Triage is a snapshot. Risk isn't.” — Closing the Gap Between Triage and Treatment

AI recommends. Safety rules protect. Clinicians decide.

Challenge	Stack	Test Verification	Repository
Accenture Challenge Round 2 Prototype	Python 3.13 / FastAPI / React 19 / HL7 FHIR	51/51 Automated Tests Passing (100%)	github.com/freya1705/PatientTriageAI

1. The Problem: What Happens After Triage?

When a patient enters the Emergency Department, nurses perform an initial triage assessment. This gives the hospital a snapshot of the patient's condition at that moment. But patients may then spend hours waiting for a bed or doctor. For example, an ESI Level 3 or 4 patient may wait around 2.5–4.5 hours. During this time, the patient's condition can change (low oxygen, increasing heart rate, sepsis, internal bleeding). The problem is that someone may not notice quickly enough that condition has changed. **PatientTriage.ai is designed to help close this gap.**

2. How PatientTriage.ai Works

- **1. Patient Risk:** How serious was the patient's condition during the original triage?
- **2. Patient Deterioration:** Are vital signs getting worse? The system tracks change over time (∆SpO■ falling, ∆HR increasing).
- **3. Data Freshness:** How recently were vital signs checked? Old information is flagged so it is not treated as fresh.
- **4. Missing Information (*Unknown does not mean safe*):** If data is missing or a monitor disconnects, the system does not assume stability. It marks the patient as uncertain and requests a physical check.

3. Four Patient Statuses

Status	What it means	What the nurse does
■ CONTINUE	Patient is stable and data is recent.	Continue monitoring in waiting lounge.
■ REASSESS	Vitals are getting old or showing moderate change (∆HR ≥ +20 bpm).	Check the patient again (targeted bedside refresh).
■ ESCALATE	Serious warning sign (SpO■ < 85%, SBP < 75 mmHg) or rapid collapse.	Immediately involve doctor / resuscitation team.
■ UNCERTAIN	Important data is missing or monitoring is disconnected.	Physically verify the patient (Unknown ≠ Safe).

4. Patient Priority Score & Operational Logic

`Priority Score = Risk + Deterioration + Old Data + Missing Data – Current Clinical Attention`

In simple terms: Higher original risk → higher priority • Patient getting worse → higher priority • Old vital signs → higher priority • Missing info → higher priority • Already attended by clinician → lower waiting-room priority.

Ranges: Base Risk (ESI w_r = 1.0) • Deterioration (+25 to +40) • Staleness (+20 to +35) • Uncertainty (+15 to +25) • Clinical Coverage (–35 when attended).

Why this matters: A normal list keeps static order. PatientTriage.ai moves deteriorating patients higher in the queue as needs change.

5. Main Dashboard & Focused Action Queue

The prototype is designed around a simple nurse workflow showing patient name/ID, current status, latest vitals, trajectory change, data age, missing data, priority rank, and recommended action. The nurse can instantly answer: **“Who do I need to check right now?”** without manually scanning a 40-patient list.

6. What Makes PatientTriage.ai Different?

- **1. It looks at change, not just the latest number:** Rapid heart rate spikes or oxygen desaturation trends matter before critical collapse.
- **2. Missing data is treated as a warning:** Unmonitored or disconnected patients trigger an UNCERTAIN state and a request for physical verification.
- **3. It focuses on patients who may be overlooked:** Clinician coverage discounting surfaces unattended waiting patients to the top.

7. Safety First: Deterministic Rules & Fail-Safe Architecture

- **Decision Support Only:** PatientTriage.ai does not replace clinicians; licensed medical professionals retain 100% decision authority.
- **Fixed Safety Thresholds:** Critical conditions (SpO■ < 85%, SBP < 75 mmHg) directly trigger an escalation without being masked by an AI score.
- **Fail-Safe Behavior:** If network connectivity or telemetry fails, the system prompts manual rounding while local alerts remain active.

8. Prototype Testing (20 Synthetic Scenarios & 51 CI/CD Tests)

Tested using 20 synthetic patient scenarios and 51 automated backend pytest tests (100% pass rate):

Test Area	Prototype Result	Verification Summary
Waiting-room deterioration detection	**20 / 20 detected**	Trajectory velocity flags silent collapse early.
Old / stale observation detection	**20 / 20 detected**	Eliminates unmonitored blind spots.
Missing vital protection	**No false reassurance**	Forces human verification via uncertainty scoring.
Patient prioritization	**Dynamic Re-Ranking**	Prioritizes unmonitored patients over attended beds.
Unsafe priority downgrade	**100% Protected**	Hard deterministic floor blocks unsafe score drops.

Important note: These are synthetic prototype test results, not results from a real hospital deployment. The next step is real-world clinical validation.

9. Technology Stack & Edge Processing

- **Frontend:** React 19 for the nurse-facing clinical cockpit • **Backend:** Python 3.13 + FastAPI for patient monitoring & prioritization logic.
- **Integration & Processing:** HL7 FHIR standard resources (Encounter, Observation); air-gapped on-premise edge processing with sub-15ms inference.

10. Future Roadmap

- **Phase 1 (Completed - Q3 2026):** Prototype developed, safety rules implemented, 51 automated tests passed.
- **Phase 2 (Q4 2026):** Hospital shadow testing alongside existing EHRs without making clinical decisions to compare recommendations.
- **Phase 3 (Q1–Q2 2027):** Live ED pilot targeting >45% reduction in Mean Time to Escalation and ≥25% reduction in Left Without Being Seen.
- **Phase 4 (Q3 2027+):** Multi-hospital expansion, ambulance telemetry, and community diversion support.

11. Quick Start & Execution

```
git clone https://github.com/freya1705/PatientTriageAI.git && cd PatientTriageAI && python -m pytest -v && .\start.ps1
```